

Sebastian Ashcallay Silva

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Software developer and AI/cloud practitioner with experience building **end-to-end AI systems, scalable web applications, and data-driven tools**. Strong background in **machine learning, multimodal pipelines, and full-stack development**, with a focus on translating ideas into real-world, user-facing products.

EDUCATION

University of Maryland College Park, MD
Bachelor of Science in Computer Science (Machine Learning Track), Minor in Mathematics Jan. 2024 – Dec. 2025

PROFESSIONAL EXPERIENCE

Software Developer Intern Jan. 2025 – Present
Apollonian — Mental Wellness Platform Remote

- Led development of a **multimodal video generation pipeline** integrating LLM-based script generation, image synthesis, text-to-speech, and automated rendering; processed 5,000+ interactions and supported 350+ daily users.
- Built an **AI-assisted CRM/outreach system** managing 50+ contacts, generating emails, LinkedIn notes, and call scripts via deterministic workflows with OpenAI Agents, while enabling manual editing and oversight.
- Implemented **Firestore Authentication** and **Firestore** for secure user management and scalable NoSQL data storage; deployed frontend using **Firestore Hosting**.
- Designed and integrated **FastAPI REST APIs** and **Python** backend services to orchestrate agentic AI workflows and content generation pipelines. Established **CI/CD pipelines** for streamlined deployment and iteration.
- Designed Firestore schemas, Cloud Storage bucket configuration and REST API workflows for structured storage of video metadata, assets, and user-generated content
- Implemented asynchronous processing pipelines using batched API requests and serverless triggers, reducing video generation runtime by ~50% (6 min → 3 min)

Tutor Sep. 2024 – Jul. 2025
C2 Education Bethesda, MD

- Led 15+ weekly sessions across Math, Physics, and Computer Science, improving student performance in standardized testing.
- Increased SAT Math scores from 580–620 to 700+ through structured learning plans and performance tracking.

PROJECTS

Professor Ratings Predictor | *Python, Pandas, Scikit-Learn, NLTK* Feb. 2025 – Apr. 2025

- Built machine learning workflow analyzing 37,000+ student reviews across 4,100 professors, transforming unstructured text into structured predictive features
- Performed sentiment analysis using NLTK and engineered statistical features correlated with instructor ratings
- Trained and evaluated regression models (e.g., Ridge, SVR) with 10-fold cross-validation, weights and boosting, achieving RMSE 0.452 and R^2 0.785
- Conducted feature importance analysis and dimensionality reduction (PCA) to interpret model behavior and generate insights from student feedback data

City Classification via CNNs | *Python, PyTorch* Mar. 2025 – May 2025

- Developed CNN classification system to predict U.S. city locations from stereographic projections (sky-based 2D mappings) using transfer learning with ResNet18, achieving 95.25% precision and 100% Top-3 accuracy
- Designed preprocessing and augmentation pipelines (noise, blur, light pollution), creating 2 datasets with 2,700+ images to improve robustness.
- Implemented full training and evaluation workflow using PyTorch and Scikit-Learn, including normalization, F1 scores, and confusion matrix analysis.
- Co-authored a 5-page report summarizing methodology, experiments, and model performance insights.

Personal Website | *React, FastAPI, Python, JavaScript, HTML, CSS* Jul. 2025 – Present

- Built a bilingual React portfolio website with blog management and dynamic file retrieval/download functionality.

RESEARCH / WRITING

- Is Generative A.I. the Future of Education?** | EdTech & AI in K-12
- Feb. 2026
- Authored analytical paper in English and Spanish examining generative AI in classrooms, focusing on learning retention, student engagement, and academic integrity.
 - Evaluated risks and proposed human-AI collaboration strategies to mitigate hallucinations and superficial learning.

CERTIFICATIONS

- AWS Cloud Practitioner**
Amazon Web Services
- 2026

TECHNICAL SKILLS

- Programming Languages:** Python, JavaScript, SQL, Java, C/C++, HTML, CSS
- AI / Machine Learning:** Machine Learning, Deep Learning, Generative AI, LLMs, Agentic AI, NLP, CNNs, Transfer Learning, Prompt Engineering, Model Evaluation, Feature Engineering, Sentiment Analysis
- Frameworks & Libraries:** FastAPI, React, Node.js, Scikit-Learn, PyTorch, NLTK
- Cloud & Backend Systems:** Google Cloud Platform (GCP), AWS, Firebase (Authentication, Firestore, Hosting, Functions, Cloud Storage), REST APIs, Serverless Architectures, Microservices
- Data Engineering & Tools:** Pandas, NumPy, NoSQL Databases, JSON, Data Pipelines, Data Modeling, ETL Concepts
- DevOps & Workflow:** Git, CI/CD Pipelines, Docker (basic), Linux/Bash, Asynchronous Processing, API Orchestration
- Other:** Bilingual (English/Spanish), Technical Writing, Product-Oriented Development, Analytical Problem Solving